

(3 Hours)

[Total Marks : 80]

- N.B. :** (1) Question No. 1 is **compulsory**.
 (2) Solve any **three** questions from **remaining**.
 (3) **Figures** to the **right** indicate **full** marks.
 (4) Assume **suitable** data if **necessary**.

1. (a) Perform following without converting into other bases. 5
 (i) $(57)_8 * (24)_8$ (ii) $(312.0)_4 + (213.2)_4$
 (b) Define following Parameters for CMOS family :— 5
 (i) Fan out (ii) Fan in.
 (c) Design a full adder using half adder and additional gates. 5
 (d) Explain concept of bistable multivibrators. 5
2. (a) Using Quine MC Clusky method determine minimal SOP form for : 10
 $F(A, B, C, D) = \sum m(1, 2, 3, 6, 7, 10, 12, 14)$
 (b) Obtain even Parity hamming code for 1010. Prove that hamming code is an error 10
 detecting and correcting code.
3. (a) Explain the operation of 4-bit universal shift register. 10
 (b) Design a 2-bit digital comparator that accepts inputs A and B and gives three outputs 10
 G, E and L.
 (i) Output G, when $A > B$
 (ii) Output E, when $A = B$
 (iii) Output L, when $A < B$.
4. (a) Implement the following using 8:1 Mux. 10
 $F(A, B, C, D) = \sum m(1, 3, 5, 9, 11, 12, 13)$.
 (b) Simplify following function using k-map. 10
 $F(A, B, C, D) = \sum m(1, 2, 3, 4, 6, 8, 10, 14, 15)$
5. (a) Design a sequence generator for following sequence. Identify and check for lock 10
 out condition
 $0 \rightarrow 3 \rightarrow 5 \rightarrow 6 \rightarrow 0$.
 (b) Explain 4-bit Johnson counter. Draw its timing diagram. 10
6. Attempt any two :— 20
 (a) Working of Master-Slave J-K flip flop.
 (b) Details and comparison of FPGA and CPLD.
 (c) Convert the following :—
 (i) SR to JK (iii) JK to D
 (ii) SR to D (iv) JK to SR.